



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (ISE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21RIE114 - SIGNAL PROCESSING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Find the 4 - point DFT of the sequence $x(n) = \cos\left(\frac{n\pi}{4}\right)$ and also plot it's magnitude & phase spectra.	10	L*3	1	1
	b) Find the circular convolution of the following two sequences using matrix method and frequency domain approach $x(n)=[1,2,3,1]$; $h(n)=[4,3,2,2]$	10	L3	1	1
2.	a) Find the output $y(n)$ of a filter whose impulse response $h(n) = \{1,1,1\}$ and input signal $x(n) = \{3,-1,0,1,3,2,0,1,2,1\}$ using overlap save method, assuming the length of the block as 5.	10	L3	2	1
	b) Find IDFT of the $x(k)=[0, 2+2j, -j4, 2-2j, 0, 2+2j, j4, 2-2j]$ using inverse radix -2 DITFFT algorithm.	10	L3	2	1
3.	a) Prove the periodicity and Linearity properties of DFT.	10	L2	1	1
	b) Find the DFT of a sequence $x(n) = \{1,1,1,1,0,0,0\}$ using DITFFT algorithm.	10	L3	2	1
Unit – II					
4.	a) A Lowpass filter is to be designed with the following desired frequency response $H_d(e^{j\omega}) = \begin{cases} e^{j\omega} & \text{for } 3\pi/4 \leq \omega \leq 3\pi/4 \\ 0 & 3\pi/4 \leq \omega \leq \pi \end{cases}$ Determine the filter coefficients of FIR filter if Hamming window is used with $N=7$.	10	L3	3	1
	b) A FIR filter is given by $y(n) = x(n) + \frac{5}{2}x(n-1) + \frac{3}{4}x(n-2) + \frac{1}{3}x(n-3)$. Draw the lattice structure.	10	L3	3	1
5.	a) Design a Butterworth digital low pass filter with following specification. (i) 3dB attenuation at passband frequency 1.5 kHz. (ii) 10dB attenuation at the frequency 3 kHz. (iii) Sampling frequency 8000 Hz.	10	L3	4	1
	b) A filter is given by the difference equation $y(n) - \frac{1}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{2}x(n-2)$. Draw the direct form I and direct form II realization.	10	L3	4	1

P.T.O.

6. a) Given a lowpass prototype $H_p(s) = \frac{1}{s+1}$.
- Determine the low and highpass filter with a cut off frequency of 40 rad/sec.
 - Determine the band pass filter with a center frequency of 100 rad/sec and bandwidth of 20 rad/sec.
- b) Explain the general procedure for IIR filter design using Bilinear transformation.
- c) Explain the Four window techniques for designing FIR filters.

6	L3	4	1
6	L3	4	1
8	L2	3	1

Unit – III

7. a) With a neat block diagram, explain the Digital signal processor based on the Harvard architecture.
- b) Discuss briefly the following special digital signal processor hardware units.
- Multiplier and accumulator
 - Shifter
 - Address Generator
8. a) Discuss the following IEEE floating point formats.
- Single precision format
 - Double precision format.
- b) Find the signed Q_{15} representation of decimal number 0.560123 and – 0.160123.

10	L2	5	1
10	L2	5	1
10	L2	5	1
10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
